

# K-Means from Scratch in Python

## Contents

|                                                               |    |
|---------------------------------------------------------------|----|
| 1. Overview . . . . .                                         | 2  |
| 2. Prerequisites . . . . .                                    | 3  |
| 3. The shape of the class . . . . .                           | 3  |
| 4. Step 2: random starting centroids . . . . .                | 4  |
| 5. Step 3: assigning clusters . . . . .                       | 5  |
| 5.1 One distance formula for any number of features . . . . . | 5  |
| 5.2 The nearest centroid . . . . .                            | 6  |
| 6. Step 4: moving the centroids . . . . .                     | 6  |
| 7. Step 5: stopping . . . . .                                 | 7  |
| 8. Testing the class . . . . .                                | 8  |
| 9. Bad random starts . . . . .                                | 10 |
| 10. Summary . . . . .                                         | 12 |
| 11. Sources . . . . .                                         | 12 |
| 12. Key terms . . . . .                                       | 12 |

Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Builds on:** Unsupervised learning ([Note 3](#)); Feature scaling ([Note 24](#)); Elbow method and WCSS ([Note 129](#)).
- **Leads to:** Hierarchical clustering ([Note 131](#)); DBSCAN ([Note 132](#)).
- **Compare with:** Hierarchical clustering ([Note 131](#)); DBSCAN ([Note 132](#)); Gaussian mixture model (GMM) ([Note 640](#)).

## 1. Overview

**Key point:** We write k-means (G-996) ourselves as a class with one public method, `fit_predict`, and two helpers: `assign_clusters` (step 3) and `move_centroids` (step 4).

Coding an algorithm from scratch is the surest test that we understand it. This Note turns the five steps of k-means (the [k-means Note](#), section 4) into about 40 lines of Python. Figure 1 shows how the steps map onto the methods of the class.

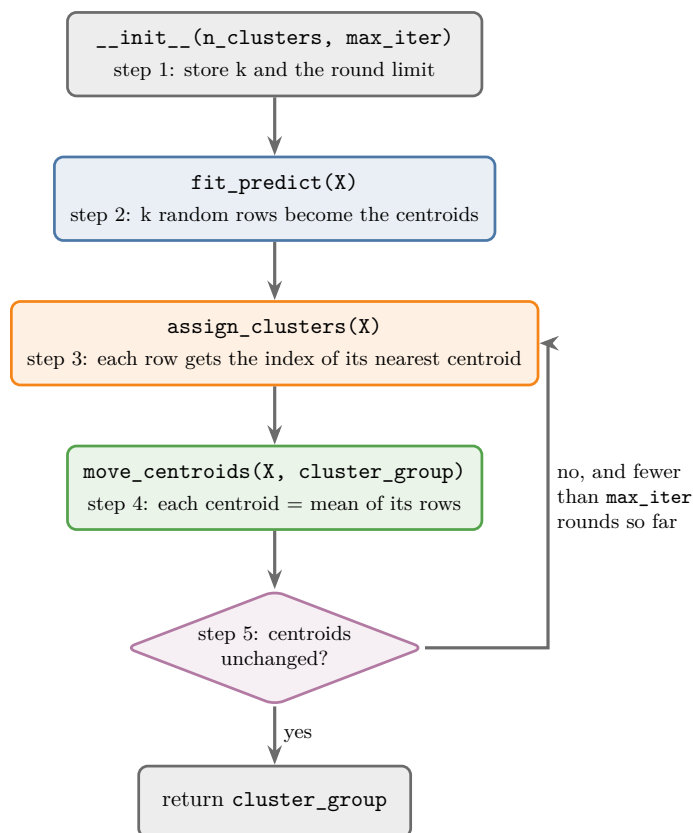


Figure 1: The KMeans class: which method does which step of k-means

The class is in `kmeans.py` in this Note's folder; running `python kmeans.py` checks that it finds the same clusters as scikit-learn. The Notebook (`notebook.ipynb`) runs every experiment below.

## 2. Prerequisites

- The five steps of k-means and WCSS (G-2102): the [k-means Note](#).
- `KMeans` in scikit-learn, the student data and boolean indexing: the [k-means in Python Note](#).

## 3. The shape of the class

**Key point:** The constructor stores `k` and a limit on the number of rounds; `fit_predict` runs the loop and returns one cluster number per observation.

We copy scikit-learn's interface, so our class is used exactly like the real one:

**Python:** Using the class.

```
from kmeans import KMeans
```

```
km = KMeans(n_clusters=4, max_iter=100)
```

```
y_means = km.fit_predict(X)    # one cluster number per row
```

`from kmeans import KMeans` loads the class from our own file `kmeans.py`, not from scikit-learn.

The **constructor** (G-457), the method that runs when the object is created, takes two settings:

- **`n_clusters`:** `k`, the number of clusters (step 1). Default 2.
- **`max_iter`:** the largest number of assign-and-move rounds. Default 100. The loop stops earlier if the centroids (G-367) stop moving.

**Python:** The constructor.

```
class KMeans:
```

```
    def __init__(self, n_clusters=2, max_iter=100,
                  random_state=None):
```

```
        self.n_clusters = n_clusters
```

```
        self.max_iter = max_iter
```

```
        self.random_state = random_state
```

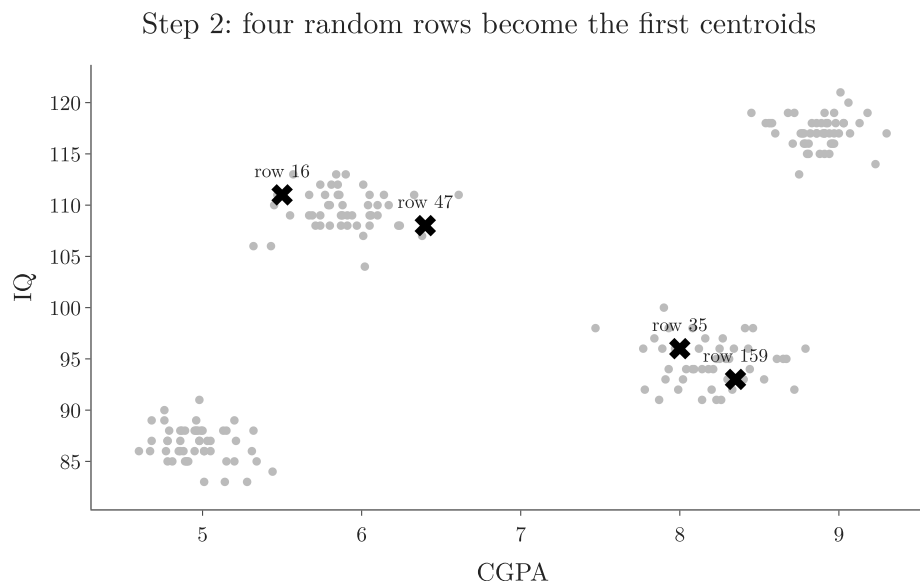
```
        self.centroids = None    # set by fit_predict
```

`random_state` is our addition: it fixes the random start so that results can be repeated.

## 4. Step 2: random starting centroids

**Key point:** Pick  $k$  different row numbers at random and take those rows as the first centroids.

$X$  is a 2-D NumPy array: with 100 points in 2 columns, it holds 100 rows of 2 numbers each. Each row is an **observation** (one record of the data) and each column a **feature** (an input variable). To start, we pick  $k$  different row numbers between 0 and 99 at random, then take those rows. Figure 2 shows one such start on the 200 students: the four chosen rows are real students. They fall in pairs inside two groups, and two groups start with no centroid at all; the loop of Figure 5 still recovers all four.



**Figure 2:** Step 2 on the students: rows 47, 16, 35 and 159, picked at random, become the first four centroids

**Python:** Choosing the starting centroids.

```
rng = np.random.default_rng(self.random_state)
random_index = rng.choice(X.shape[0],
                          size=self.n_clusters,
                          replace=False)
self.centroids = X[random_index].astype(float)
```

$X.shape[0]$  is the number of rows. `rng.choice(100, size=2, replace=False)` returns 2 different numbers from 0 to 99, for example [26, 17]. `X[[26, 17]]` then picks rows 26 and 17. `astype(float)` lets the centroids hold decimals once they move.

**Extra:** The same can be written with Python's `random` module: `random.sample(range(X.shape[0]), k)` also returns  $k$  different numbers. NumPy's generator is used here because it takes a seed per object, so two models never share

random state.

## 5. Step 3: assigning clusters

**Key point:** For every row, compute its distance to every centroid and record the index of the smallest one.

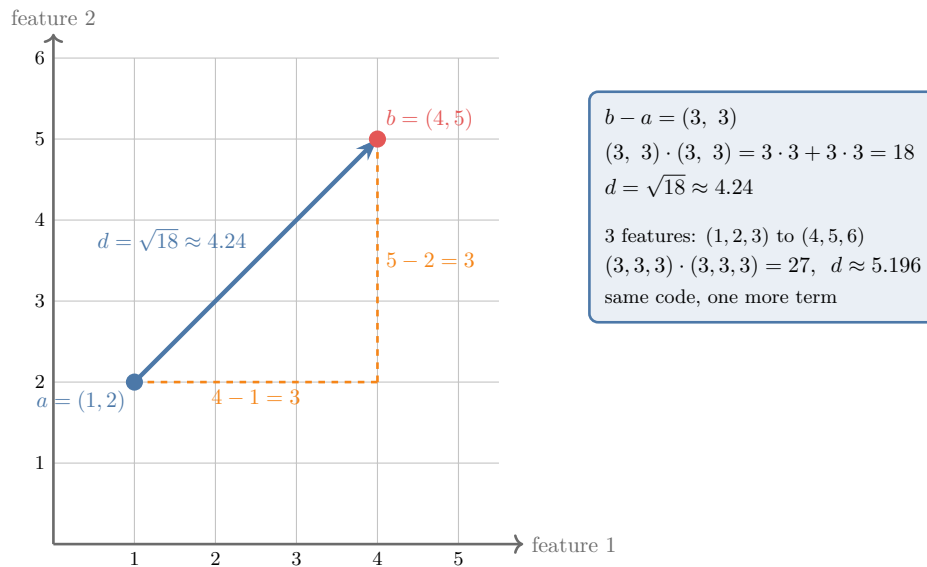
### 5.1 One distance formula for any number of features

**Key point:** The squared Euclidean distance (G-715) is the dot product (G-634) of the difference vector with itself, so `np.sqrt(np.dot(a - b, a - b))` works for 2, 3 or 100 features.

The Euclidean distance (the [KNN imputer Note](#), section 4.1) gains one term per feature, so code written term by term would change with the number of features. The sum of squared differences is the dot product (the [PCA step-by-step Note](#)) of the difference vector with itself, which needs no change:

$$d(a, b) = \sqrt{(b - a) \cdot (b - a)}$$

For  $a = (1, 2)$ ,  $b = (4, 5)$ :  $b - a = (3, 3)$ ,  $(3, 3) \cdot (3, 3) = 18$  and  $d = \sqrt{18} \approx 4.24$ . With a third feature,  $(1, 2, 3)$  and  $(4, 5, 6)$ , the dot product is 27 and  $d = \sqrt{27} \approx 5.196$ , from the same code. In Figure 3, watch the two dashed sides: each feature adds one squared side, and the dot product adds them all at once.



**Figure 3:** The distance from  $(1, 2)$  to  $(4, 5)$ : the difference vector  $(3, 3)$ , its dot product with itself, and its square root

**Python:** The distance in NumPy.

```
np.sqrt(np.dot(b - a, b - a))    # 4.24 for (1, 2), (4, 5)
```

$b - a$  subtracts element by element. `np.dot(u, v)` multiplies matching elements and adds them up.

## 5.2 The nearest centroid

**Key point:** With 100 rows and 2 centroids we compute 200 distances; each row gets the index of its smaller one.

A nested loop visits every row and, inside, every centroid. With 100 rows and 2 centroids the inner line runs  $100 \times 2 = 200$  times. For each row we keep the position of the smallest distance: 0 if the first centroid is nearer, 1 if the second.

**Python:** `assign_clusters`.

```
def assign_clusters(self, X):
    cluster_group = []
    for row in X:
        distances = [np.sqrt(np.dot(row - c, row - c))
                     for c in self.centroids]
        cluster_group.append(int(np.argmin(distances)))
    return np.array(cluster_group)
```

The square brackets build a list of  $k$  distances, one per centroid. `np.argmin` returns the position of the smallest value, the same as `distances.index(min(distances))`. The result has one entry per row, such as `[0, 0, 1, ...]`, and is returned as a NumPy array so it can be used for boolean indexing.

## 6. Step 4: moving the centroids

**Key point:** For each cluster  $k$ , select its rows with `X[cluster_group == k]` and take the mean of each column with `mean(axis=0)`.

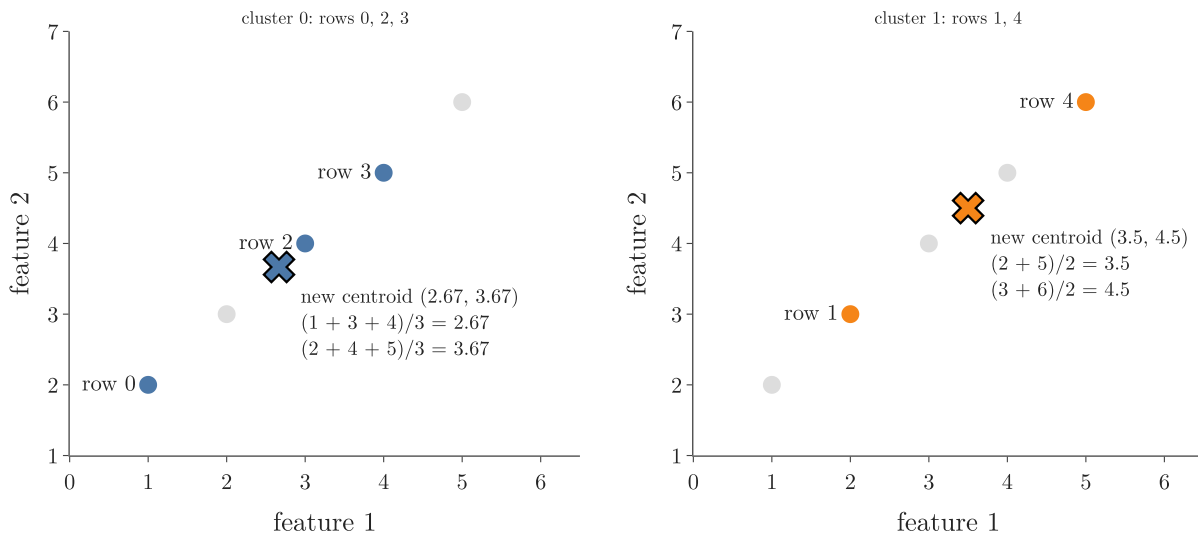
The cluster numbers tell us which rows belong to which cluster. The new centroid of a cluster is the mean of each column over its rows (the [k-means Note](#), section 4.3).

Take five rows and their clusters:

| Row | Values | Cluster |
|-----|--------|---------|
| 0   | (1, 2) | 0       |
| 1   | (2, 3) | 1       |
| 2   | (3, 4) | 0       |
| 3   | (4, 5) | 0       |
| 4   | (5, 6) | 1       |

`X[cluster_group == 0]` keeps rows 0, 2 and 3. Their column means are  $(1 + 3 + 4)/3 = 2.67$  and  $(2 + 4 + 5)/3 = 3.67$ , so the new centroid of cluster 0 is (2.67, 3.67). Cluster 1 gets

(3.5, 4.5). Figure 4 draws both: each cross lands in the middle of its own rows, and the other cluster's rows (grey) play no part.



**Figure 4:** Moving the centroids for the five-row example: each new centroid is the column mean of its own rows

**Python:** `move_centroids`.

```
def move_centroids(self, X, cluster_group):
    new_centroids = []
    for k in range(self.n_clusters):
        members = X[cluster_group == k]
        new_centroids.append(members.mean(axis=0))
    return np.array(new_centroids)
```

`axis=0` averages down each column, giving one mean per column. Without it, `mean()` averages all six numbers into a single value (3.17), which is not a point at all.

**Extra:** A cluster can end up empty: no row is nearest to its centroid. The mean of zero rows is undefined, and a loop over the cluster numbers that actually occur (for example with `np.unique(cluster_group)`) would silently return fewer than `k` centroids. The file `kmeans.py` keeps an empty cluster's old centroid instead. `scikit-learn` moves it to the data point that is farthest from its own centroid (`scikit-learn` source, `_k_means_common.pyx`).

## 7. Step 5: stopping

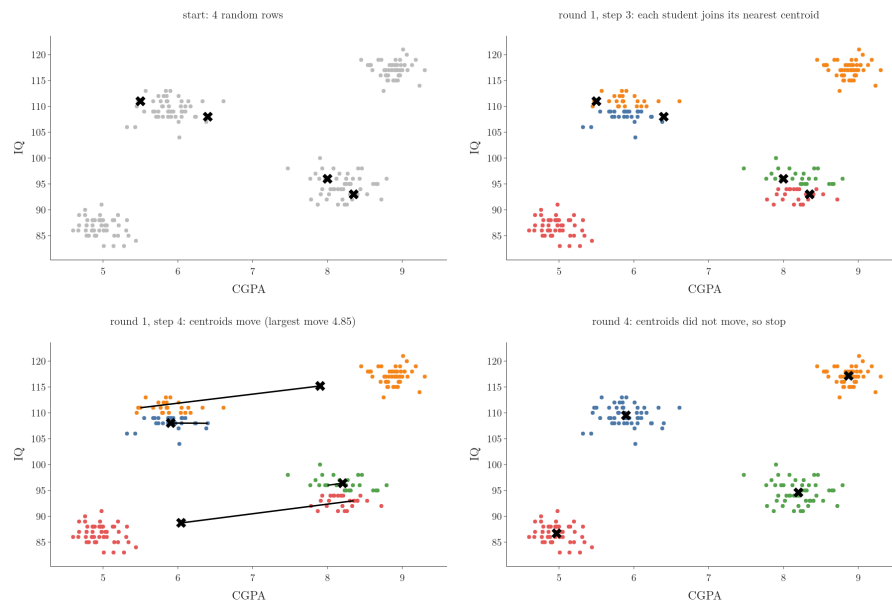
**Key point:** Save the centroids before moving them; if the new ones equal the old ones, break out of the loop.

**Python:** The loop inside `fit_predict`.

```
for i in range(self.max_iter):
    cluster_group = self.assign_clusters(X)
    old_centroids = self.centroids
    self.centroids = self.move_centroids(X, cluster_group)
    if np.allclose(old_centroids, self.centroids):
        break
return cluster_group
```

`break` leaves the loop at once. `np.allclose` is `True` when every pair of numbers is equal up to a tiny rounding error. `(old == new).all()` also works, but computed means can differ in the last decimal place.

Figure 5 runs the whole loop on the 200 students from the start of Figure 2. Watch the largest move shrink round by round; in round 4 the centroids do not move at all, `np.allclose` is `True`, and the loop breaks.



**Figure 5:** The assign-and-move loop on the students: colours change at step 3, crosses move at step 4, and the loop stops in round 4

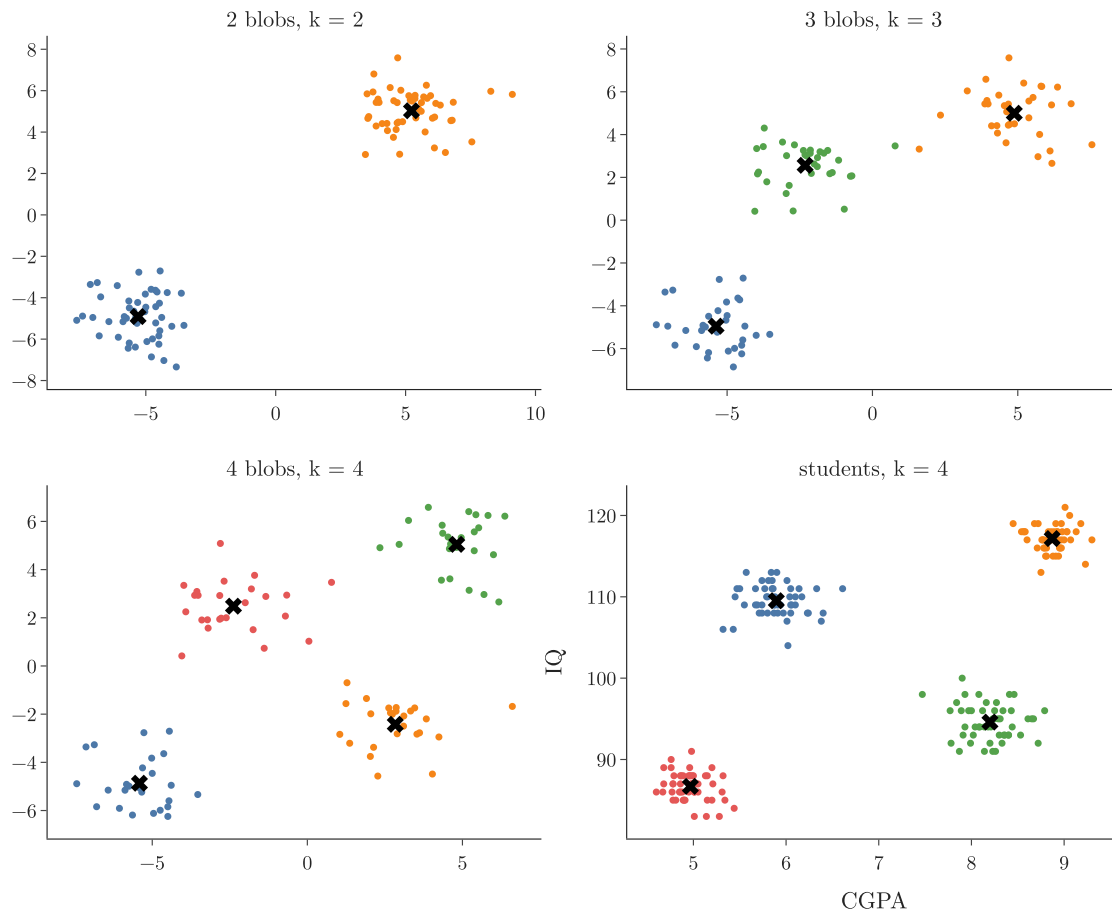
The loop ends in one of two ways (Figure 1): the centroids stop moving, or `max_iter` rounds have run. Either way, `fit_predict` returns the cluster number of every row.

## 8. Testing the class

**Key point:** On two, three and four blobs, and on the 200 students, the class finds the right groups in 2 to 5 rounds.

We generate test data with `make_blobs` (the [k-means in Python Note](#), section 8): 100 points

around 2, 3 or 4 centres, each with spread 1. Then we run our class with the matching  $k$ , and finally on the 200 students with  $k = 4$ .



**Figure 6:** The from-scratch class on 2, 3 and 4 blobs and on the students; crosses are the final centroids

Figure 6 shows all four results: every group found, every centroid in the middle of its group. The loops finished after 2, 5, 3 and 4 rounds. Plotting needs one scatter call per cluster, so the plotting code (not the class) must be extended each time  $k$  grows.

**Python:** The self-check in `kmeans.py`.

```
from sklearn.metrics import adjusted_rand_score
```

```
ours = KMeans(n_clusters=3, random_state=1).fit_predict(X)
theirs = SkKMeans(n_clusters=3, random_state=0).fit_predict(X)
assert adjusted_rand_score(ours, theirs) == 1.0
```

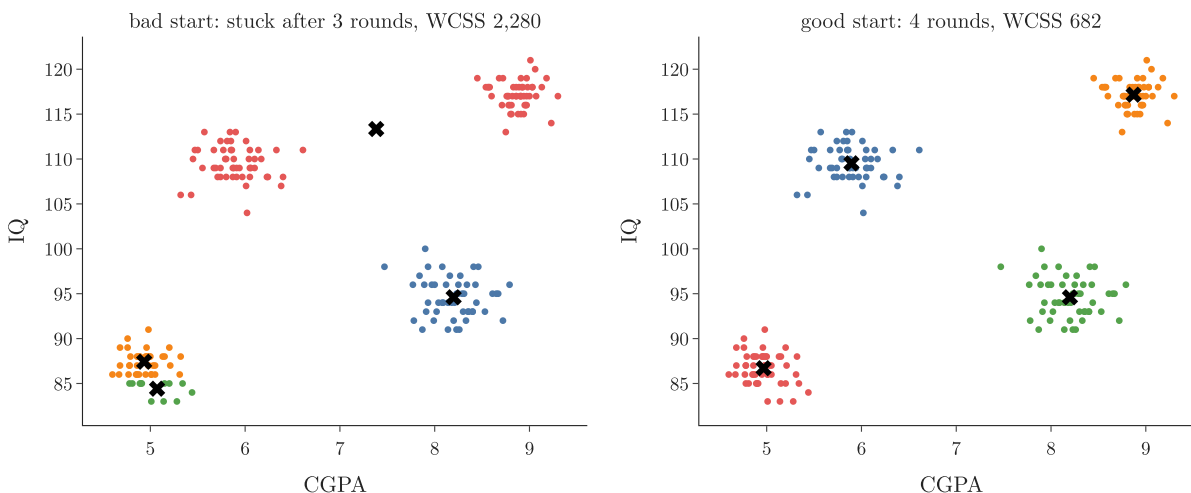
Cluster numbers are only names, so we cannot compare them directly. `adjusted_rand_score` is 1.0 when two labelings put exactly the same points together, whatever the numbers.

## 9. Bad random starts

**Key point:** If the starting centroids fall badly, k-means gets stuck in poor clusters, a **local optimum** (G-1111); more rounds do not help, but restarting from other random points does.

Sometimes the class returns wrong clusters. Blaming `max_iter` and raising it is tempting, but the round counts say otherwise. Across 30 random starts on the four blobs, k-means never needed more than 10 rounds; at most 10 of the 100 allowed rounds were ever used.

Figure 7 shows the real cause on the students. With one random start (left), two centroids end up in the bottom-left group. That group gets split in two, while two real groups, the top ones, have to share one centroid placed in the empty space between them. After 3 rounds nothing changes any more, so the algorithm stops, stuck. Raising `max_iter` to 500 gives exactly the same result.



**Figure 7:** The same students and  $k = 4$  from two random starts: stuck in poor clusters (left) and the right clusters (right)

The two runs can be compared by their WCSS (the [k-means Note](#), section 5.1): 2,280 for the bad start and 682 for the good one. Lower is better, so the fix is simple (ESL §14.3.6):

1. run k-means from several random starts;
2. keep the run with the lowest WCSS.

In the Notebook, the best of 10 starts is the good clustering (WCSS 682). `scikit-learn`'s `n_init` (the [k-means in Python Note](#), section 5) does exactly this job, and its `k-means++` start spreads the first centroids out.

**Extra:** The class also stores the WCSS of its result as `inertia_`, like `scikit-learn`:  
`self.inertia_ = ((X - self.centroids[cluster_group]) ** 2).sum()`  
`self.centroids[cluster_group]` repeats, for every row, the centroid of its own cluster;

subtracting, squaring and summing gives the WCSS. With it, the elbow method takes one loop over  $k$ . Taking the best of 10 starts for each  $k$ , the students give WCSS 29,958, 4,184, 2,363, 682, 514, ...: the elbow is at  $k = 4$ , as with scikit-learn.

## 10. Summary

| Method                            | Step | What it does                                                                                    |
|-----------------------------------|------|-------------------------------------------------------------------------------------------------|
| <code>__init__</code>             | 1    | Stores <code>k</code> ( <code>n_clusters</code> ) and the round limit ( <code>max_iter</code> ) |
| <code>fit_predict</code>          | 2    | Picks <code>k</code> random rows as the first centroids, then runs the loop                     |
| <code>assign_clusters</code>      | 3    | Distance from every row to every centroid; keeps the index of the nearest                       |
| <code>move_centroids</code>       | 4    | Each centroid becomes the column means of its rows                                              |
| check in <code>fit_predict</code> | 5    | Stops when the centroids no longer move                                                         |

- The Euclidean distance for any number of features: `np.sqrt(np.dot(a - b, a - b))`.
- `X[cluster_group == k].mean(axis=0)` is the new centroid of cluster `k`.
- On the four blobs, k-means stopped within 10 rounds from all 30 starts; the wrong result on the students came from a bad random start, not from too few rounds.
- Fix bad starts by restarting several times and keeping the lowest WCSS.

## 11. Sources

### Built from

- CampusX, “K-Means Clustering Algorithm From Scratch In Python | ML Algorithms From Scratch”, YouTube, <https://www.youtube.com/watch?v=MFraC1JObUo>

### Other references

- ESL: Hastie, Tibshirani and Friedman, *The Elements of Statistical Learning*, 2nd ed., Springer, 2009, §14.3.6 (k-means; restarting from several random starts and keeping the lowest objective).
- scikit-learn source, `_k_means_common.pyx`: the function `_relocate_empty_clusters_dense` in `sklearn/cluster/_k_means_common.pyx`, scikit-learn 1.9.

## 12. Key terms

| Term                                  | Meaning                                                                |
|---------------------------------------|------------------------------------------------------------------------|
| Constructor ( <code>__init__</code> ) | The method that runs when an object is created and stores its settings |
| <code>max_iter</code>                 | The largest number of assign-and-move rounds k-means may run           |

| Term                      | Meaning                                                                                          |
|---------------------------|--------------------------------------------------------------------------------------------------|
| <code>np.argmin</code>    | NumPy function returning the position of the smallest value                                      |
| <code>mean(axis=0)</code> | The mean of each column of an array                                                              |
| Local optimum (k-means)   | A clustering where k-means has stopped but a better one exists, caused by a bad start            |
| Adjusted Rand score       | A number that is 1.0 when two labelings group the points identically, whatever the label numbers |